WALCHAND COLLEGE OF ENGINEERING, SANGLI

(Government Aided Autonomous Institute)

WLUG
COMMUNITY | KNOWLEDGE | SHARE

Walchand Linux Users' Group

OPEN SOURCE DAY

S1

BREAKING THE MONOLITH

REGISTER HERE

S2

GIT IN FOCUS

SPONSORED BY

EUCLIDEUM

CONNECT WITH US

25 OCT

Mini CCF

FOR ANY QUERY

9881178111
8421114732

Bhargav Joshi	Dr. A. R. Surve	Dr. U. B. Chavan	Dr. A. J. Umbarkar	Prof. K. S. Khandagale	Dr. S. U. Sapkal	Dr. S. S. Rathod
President Walchand Linux Users' Group	HoD Computer Science and Engineering	HoD Information Technology	Staff Advisor Walchand Linux Users' Group	Staff Advisor Walchand College of Engineering	Dean of Academics Walchand College of Engineering	Director Walchand College of Engineering

WLUG
COMMUNITY | KNOWLEDGE | SHARE

TABLE OF CONTENT

- Version Control System
- Git and Git Architecture
- Git VS GitHub
- Basic Linux Commands

WLUG
COMMUNITY | KNOWLEDGE | SHARE

VERSION CONTROL SYSTEM

WHAT IS VCS?

- A tool to record changes to your files
- Keeps full history of your project
- Lets you revert to any earlier version
- Lets teams work on same code safely

WLUG
COMMUNITY | KNOWLEDGE | SHARE

WLUG
COMMUNITY | KNOWLEDGE | SHARE

IMPORTANCE OF VCS

- Back Ups
- Error Handling
- Maintaining Project Logs

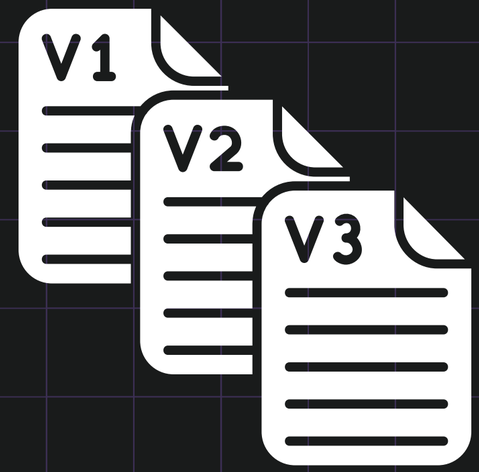
WLUG
COMMUNITY | KNOWLEDGE | SHARE

FEATURES OF VCS

WLUG
COMMUNITY | KNOWLEDGE | SHARE

MANAGING VERSIONS

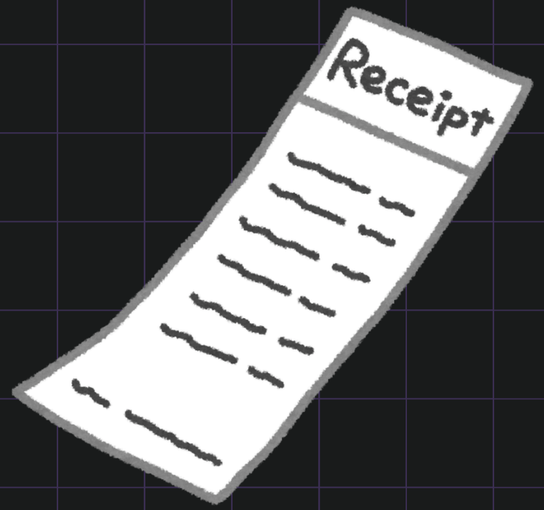
- Saves every change as a version
- Allows you to go back to earlier state
- Branches keep new work separate
- Lets you compare any two versions



WLUG
COMMUNITY | KNOWLEDGE | SHARE

AUDITING CHANGES

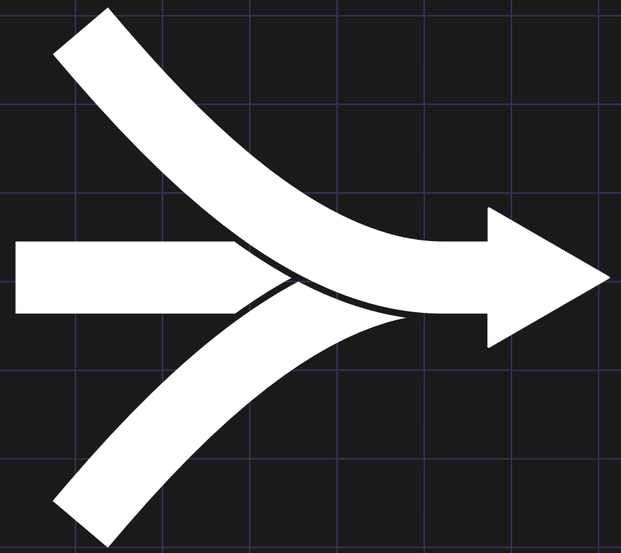
- See who made each change
- See what changed, line by line
- See when each change was made
- See why change was made



WLUG
COMMUNITY | KNOWLEDGE | SHARE

SIMPLIFY INTEGRATION

- Many people can work on the same project at the same time
- Changes to different parts merge automatically
- Same-line edits are flagged as conflicts



WLUG

COMMUNITY | KNOWLEDGE | SHARE

WLUG
COMMUNITY | KNOWLEDGE | SHARE

GIT

WLUG
COMMUNITY | KNOWLEDGE | SHARE

HISTORY OF GIT

September, 1991 — Linus Torvalds releases Linux 0.01

February, 2002 — Linus moves to BitKeeper

April, 2005 — BitKeeper access is revoked

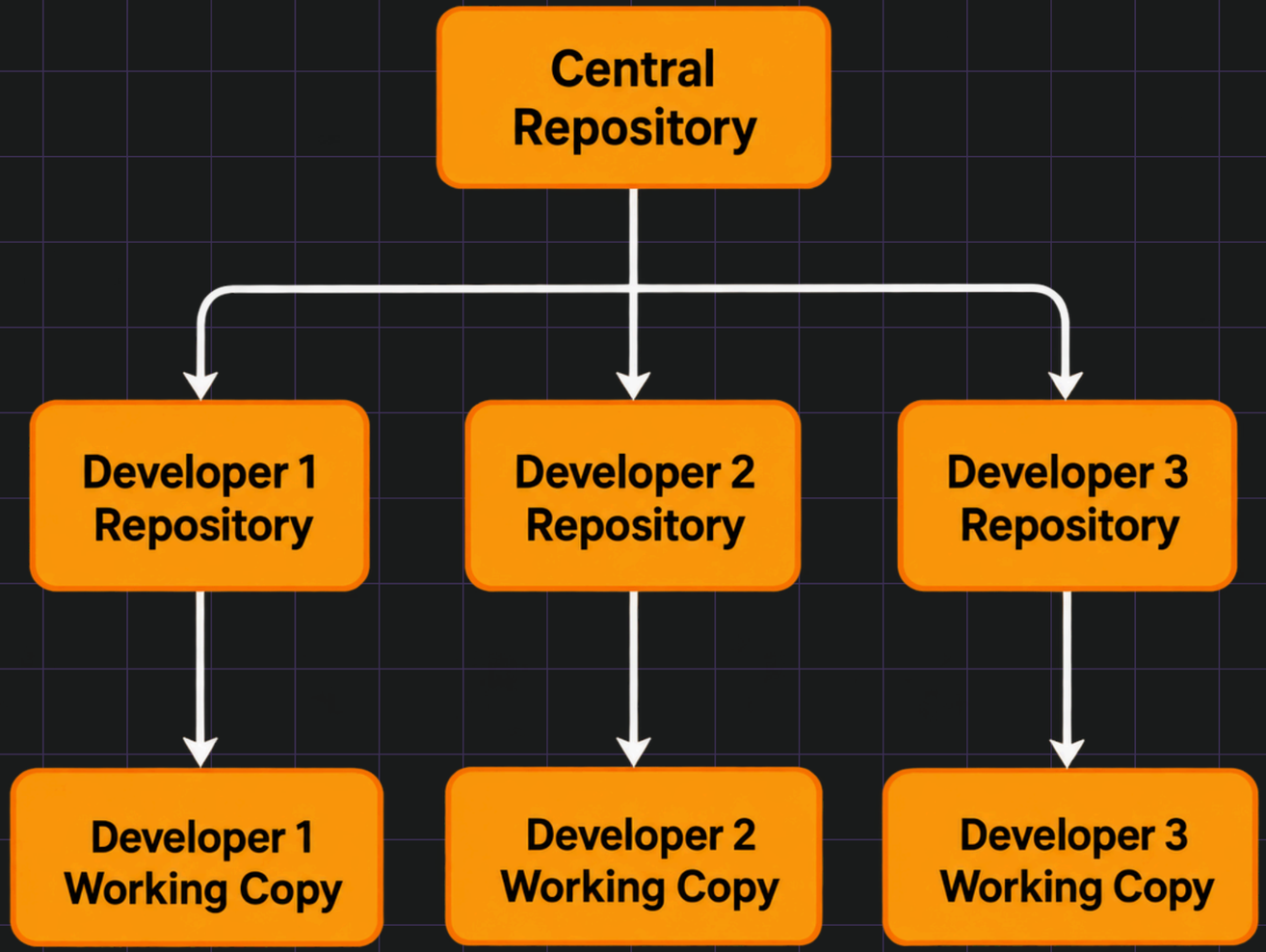
April, 2005 — Linus creates Git

July, 2005 — Junio C Hamano takes over as maintainer

September, 2026 — Git version 2.56.0 is released

FEATURES OF GIT

- Speed
- Security
- Distributed



Distributed Version Control System

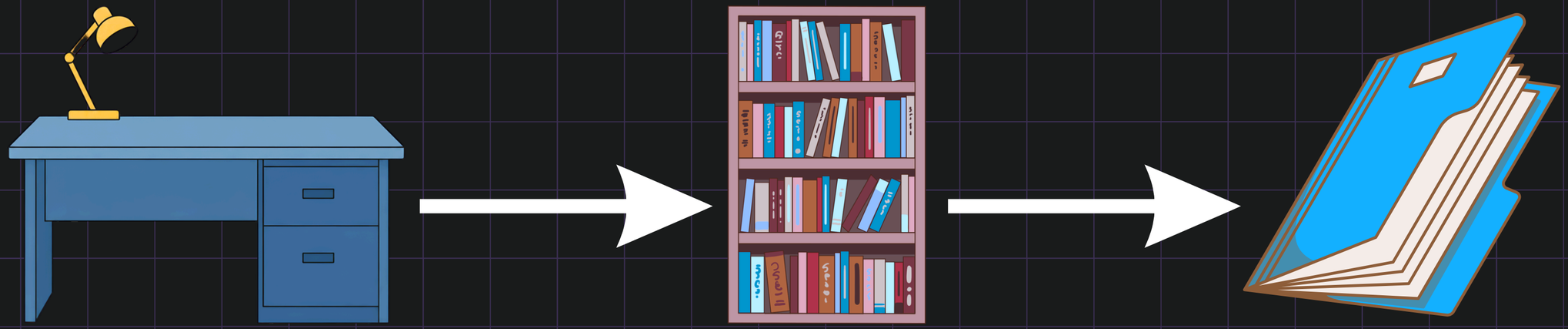
WLUG
COMMUNITY | KNOWLEDGE | SHARE

GIT ARCHITECTURE

- Working directory
- Staging Index
- Local Repository

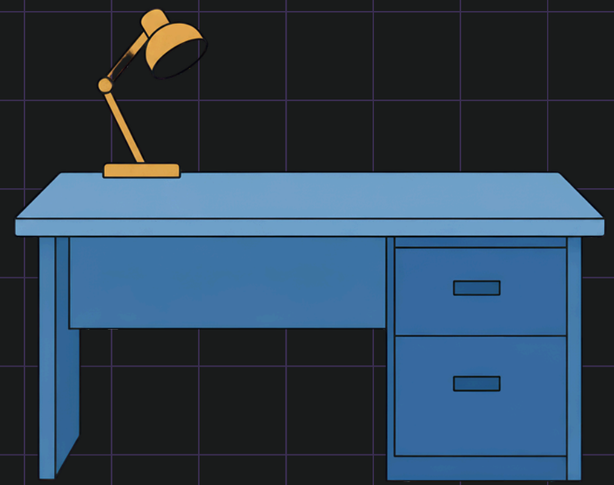
WLUG
COMMUNITY | KNOWLEDGE | SHARE

GIT ARCHITECTURE



WORKING DIRECTORY

- The project folder you open on your computer
- This is where you create, open and edit files
- Nothing here is saved to your history until you choose to save it



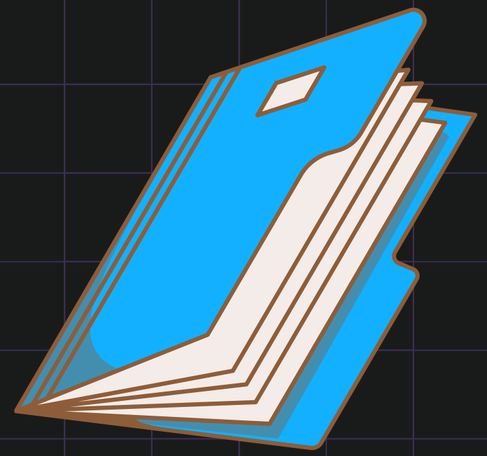
STAGING AREA

- Holding area for the changes you want in your next commit
- You can choose what changes are saved
- Lets you group related changes in a single commit

WLUG
COMMUNITY | KNOWLEDGE | SHARE

LOCAL REPOSITORY

- Stores full history of commits on your computer
- A snapshot is saved here after each commit
- It is stored locally on your computer and needs no Internet

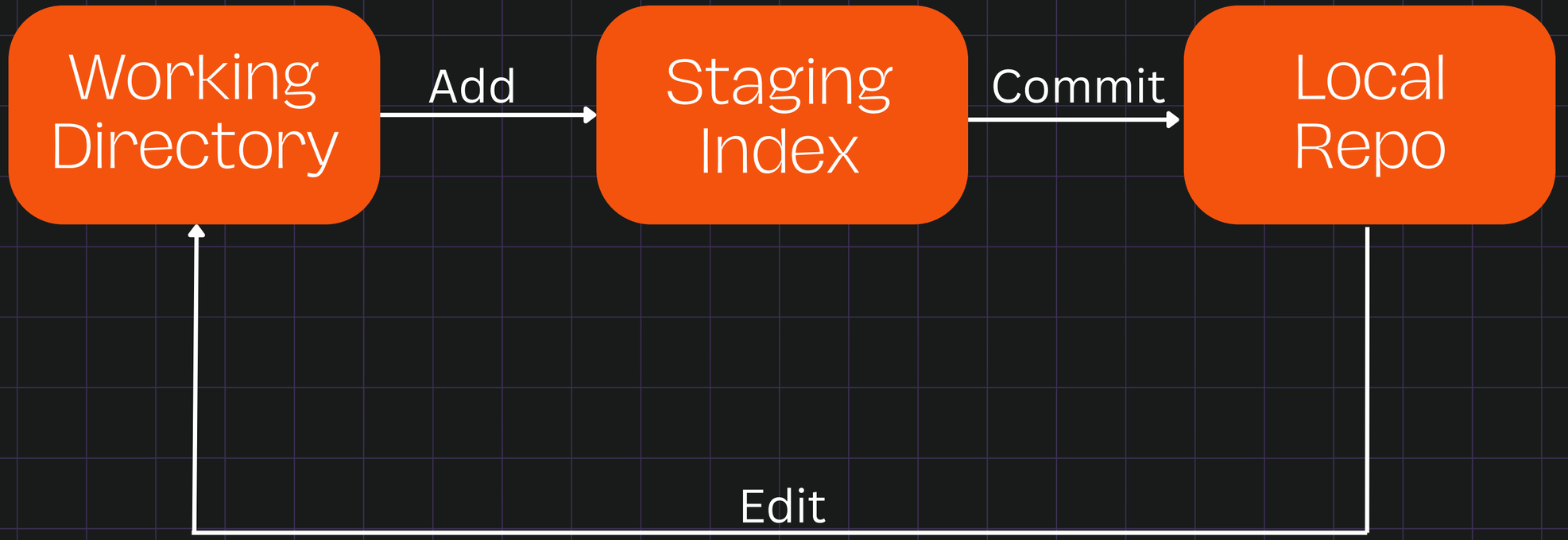


FILE STATES

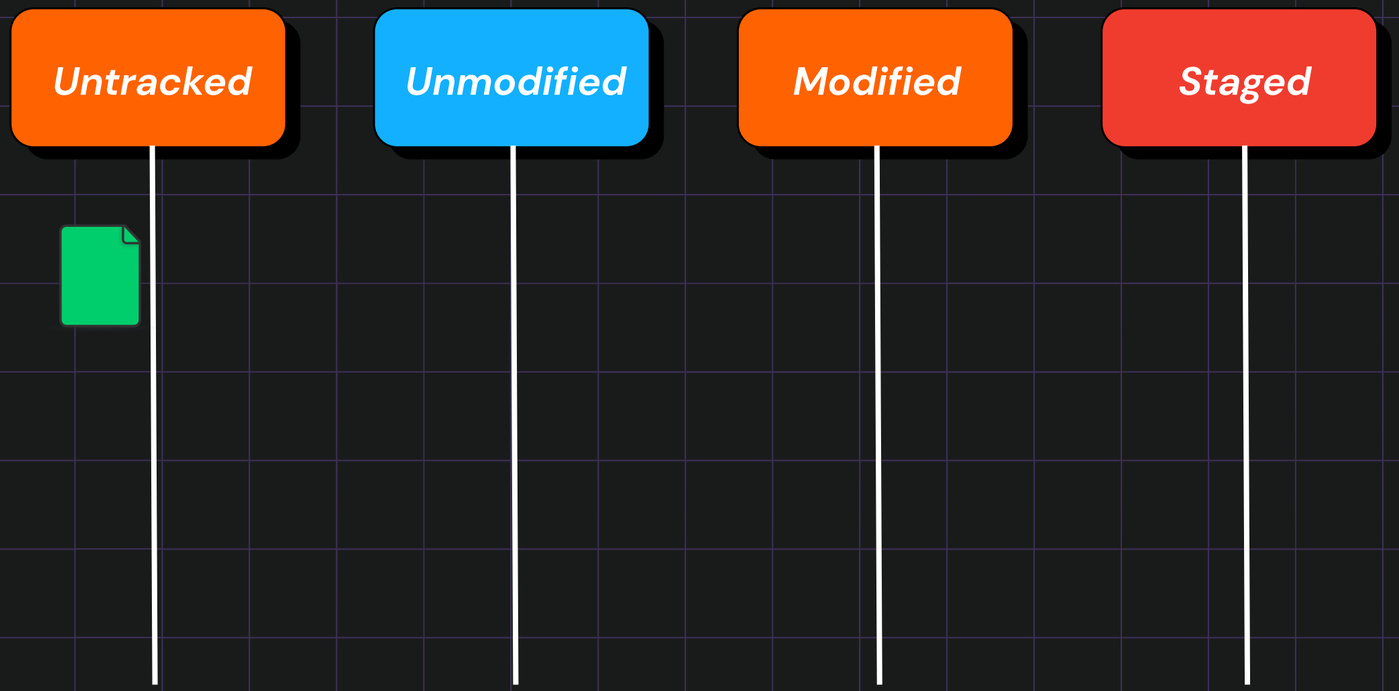
- Untracked
- Staged
- Unmodified
- Modified

WLUG
COMMUNITY | KNOWLEDGE | SHARE

GIT ARCHITECTURE



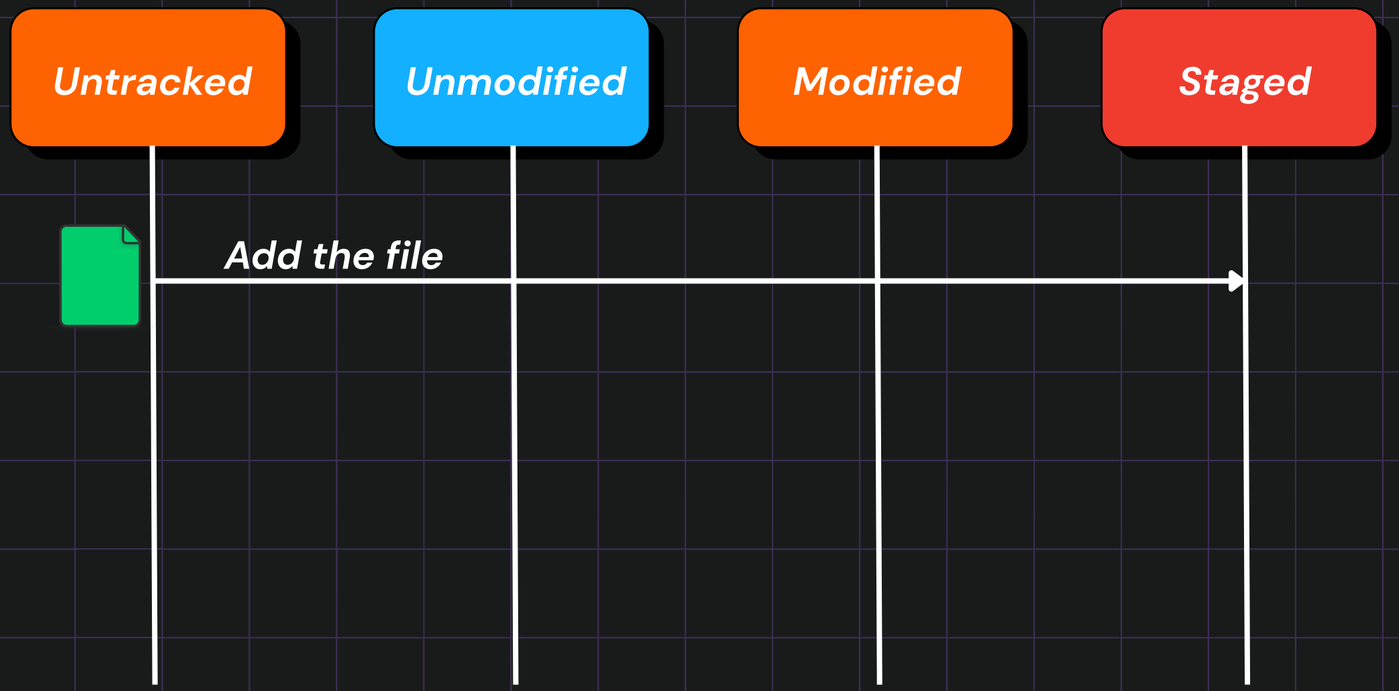
FILE STATES



File in VSCode

File in Git

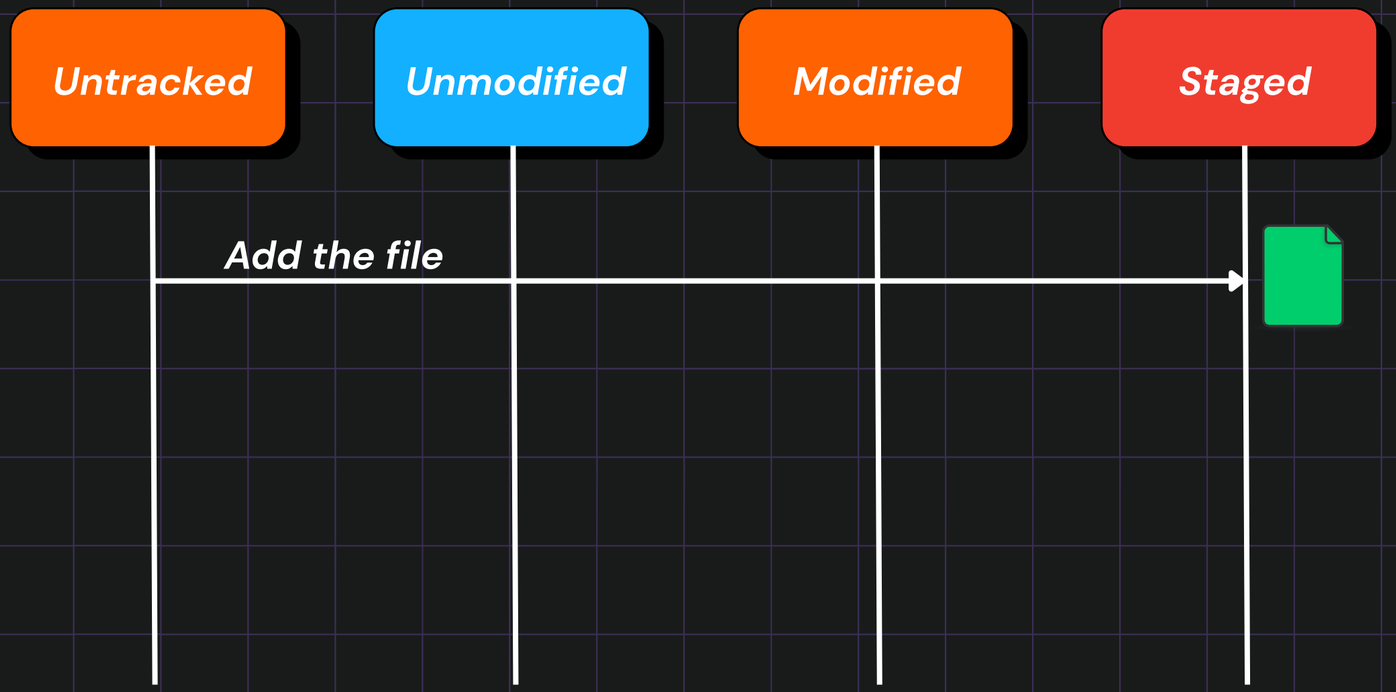
FILE STATES



File in VSCode

File in Git

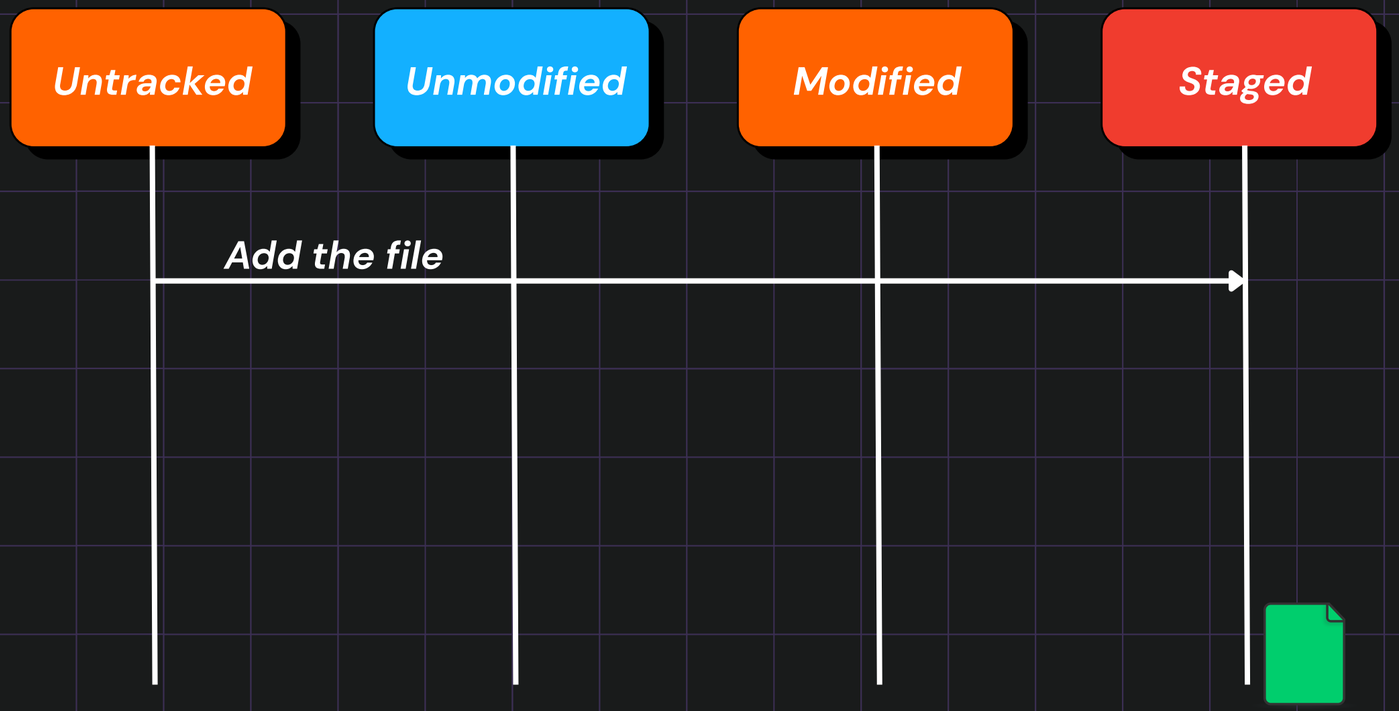
FILE STATES



File in VSCode

File in Git

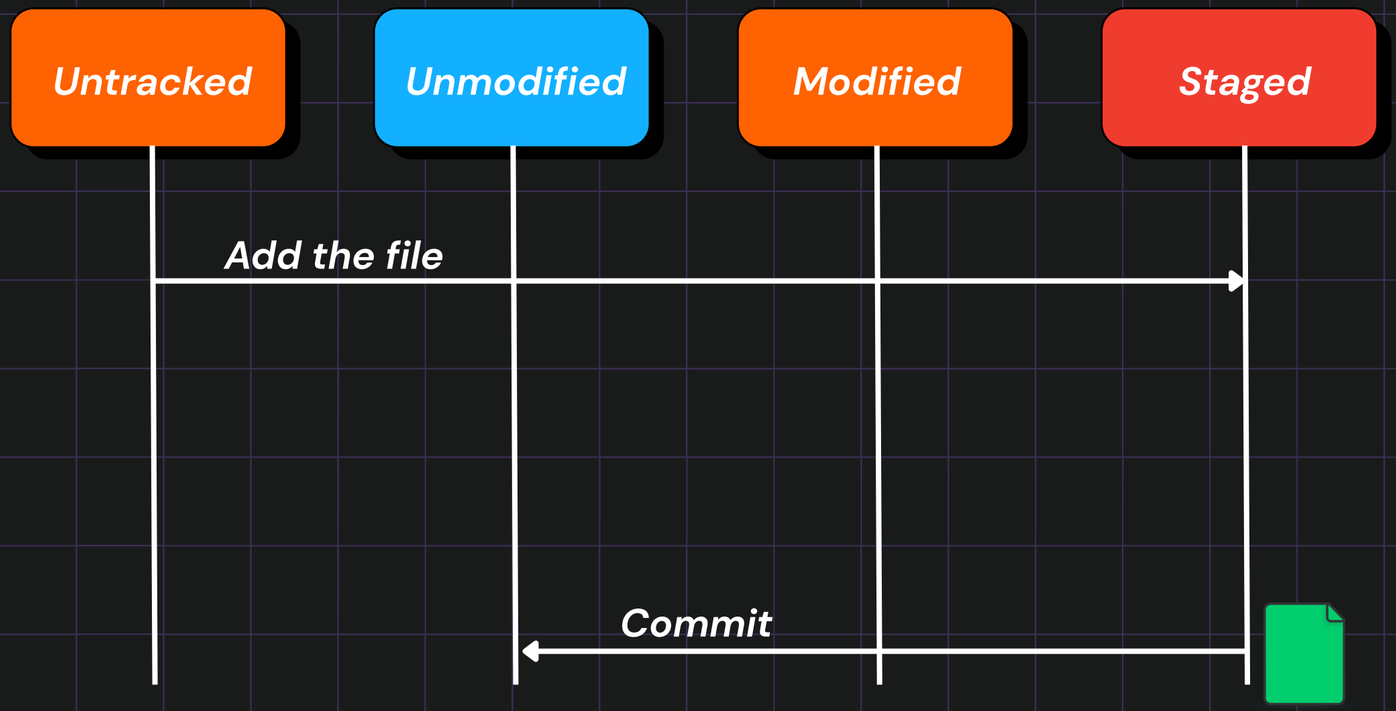
FILE STATES



File in VSCode

File in Git

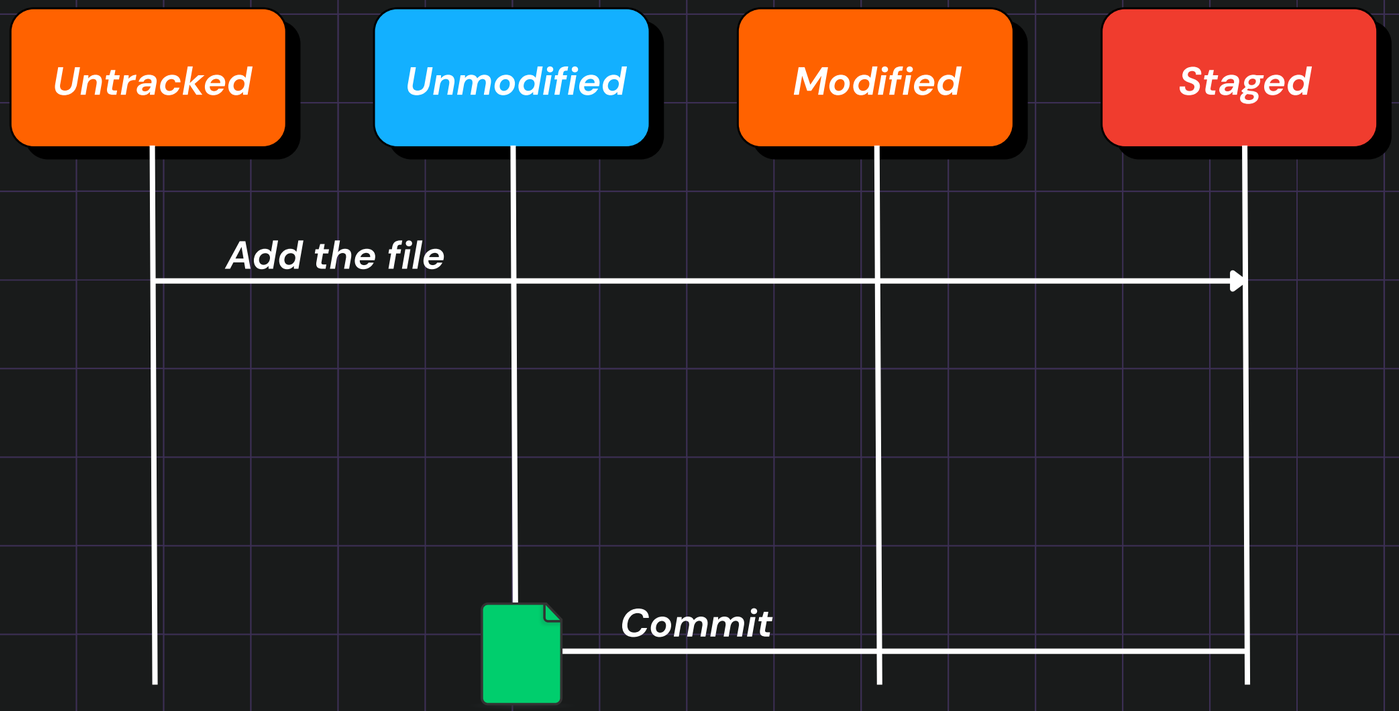
FILE STATES



File in VSCode

File in Git

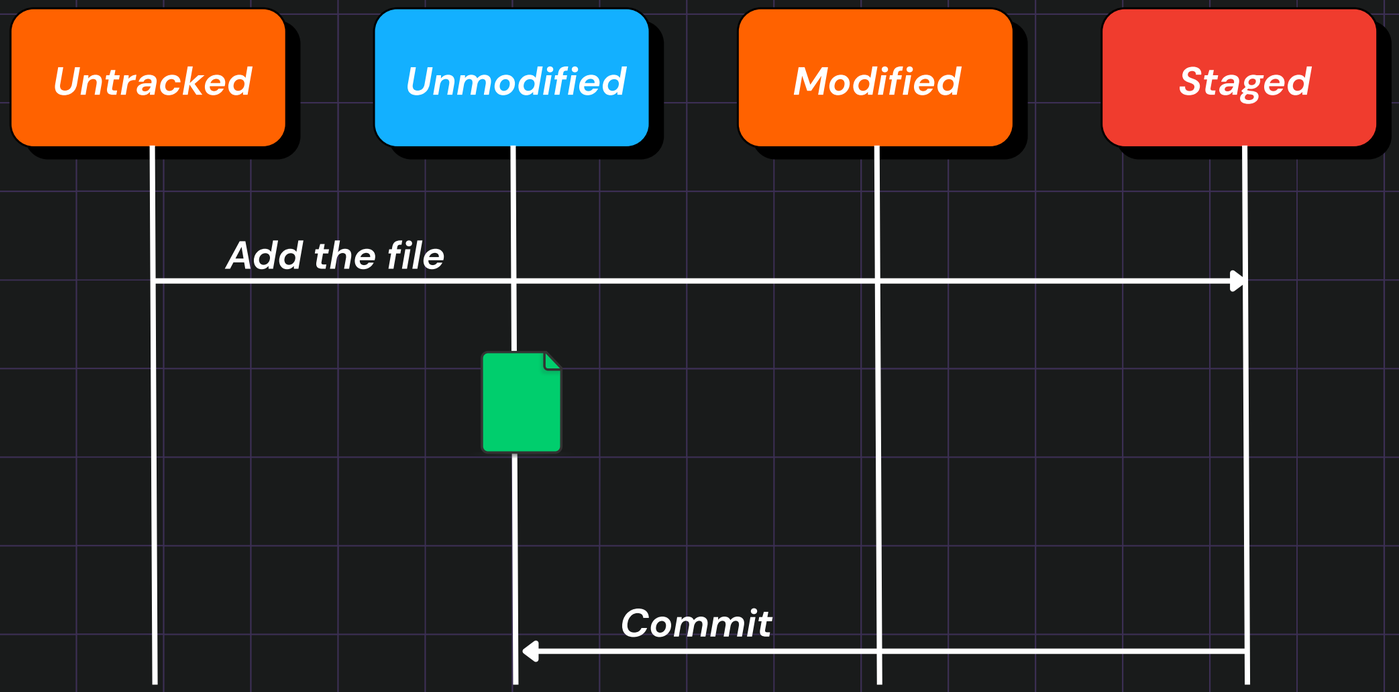
FILE STATES



File in VSCode

File in Git

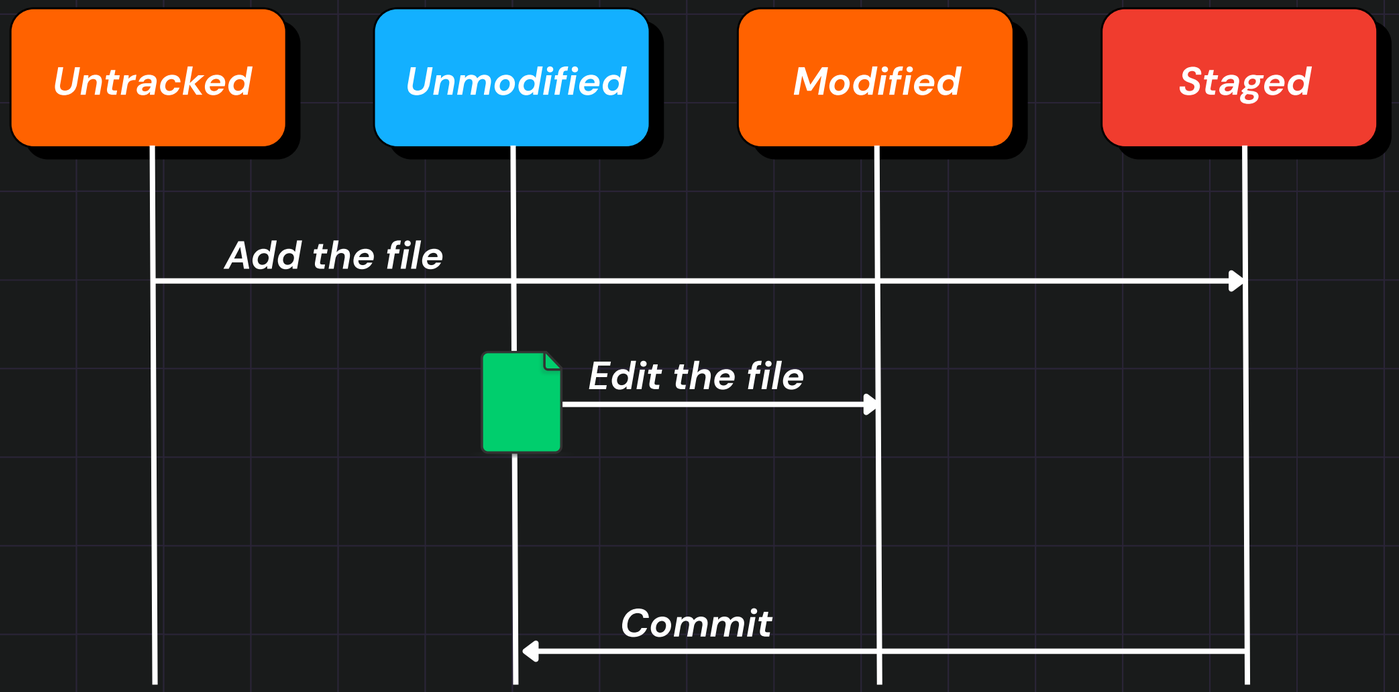
FILE STATES



File in VSCode

File in Git

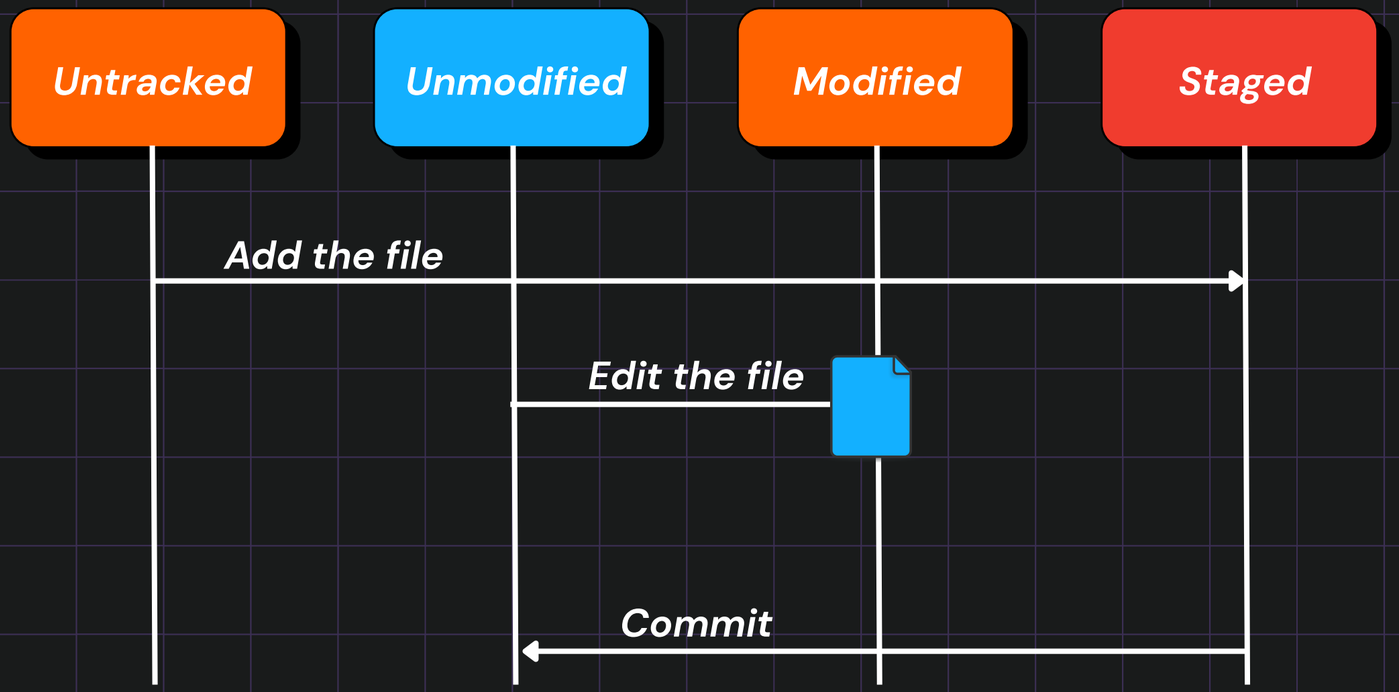
FILE STATES



File in VSCode

File in Git

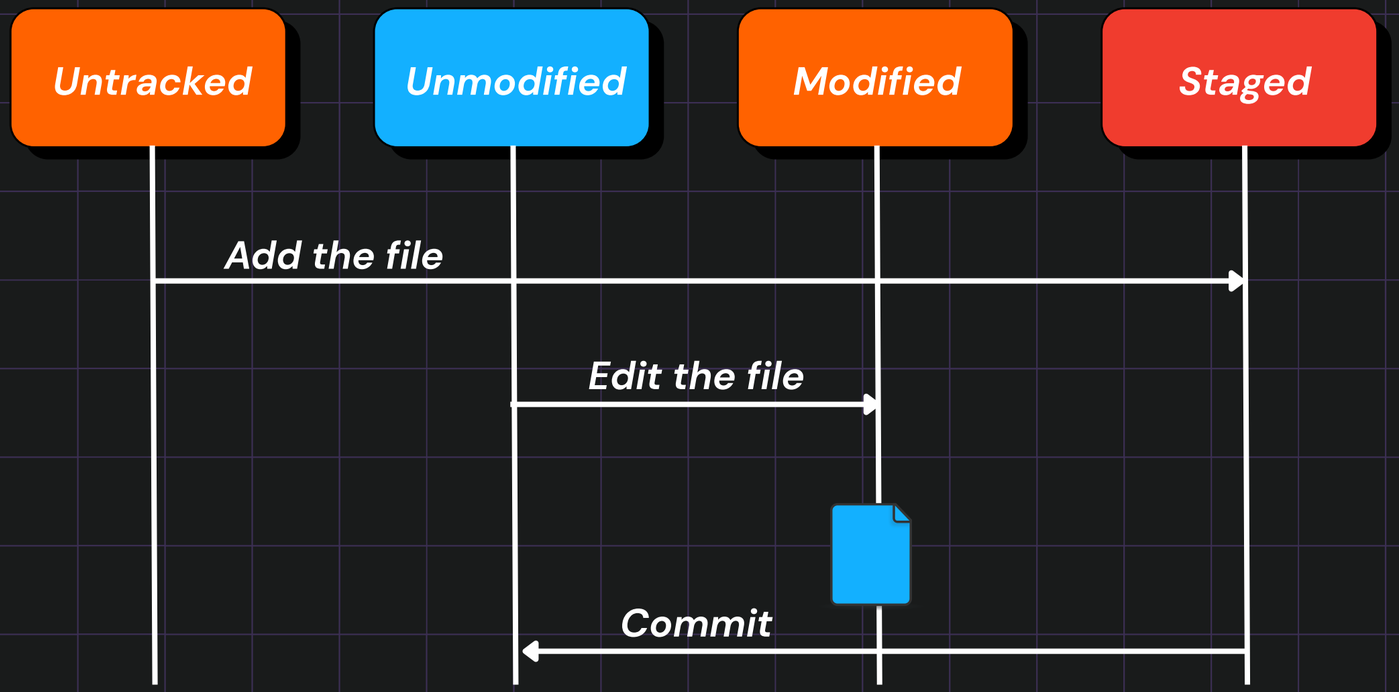
FILE STATES



File in VSCode

File in Git

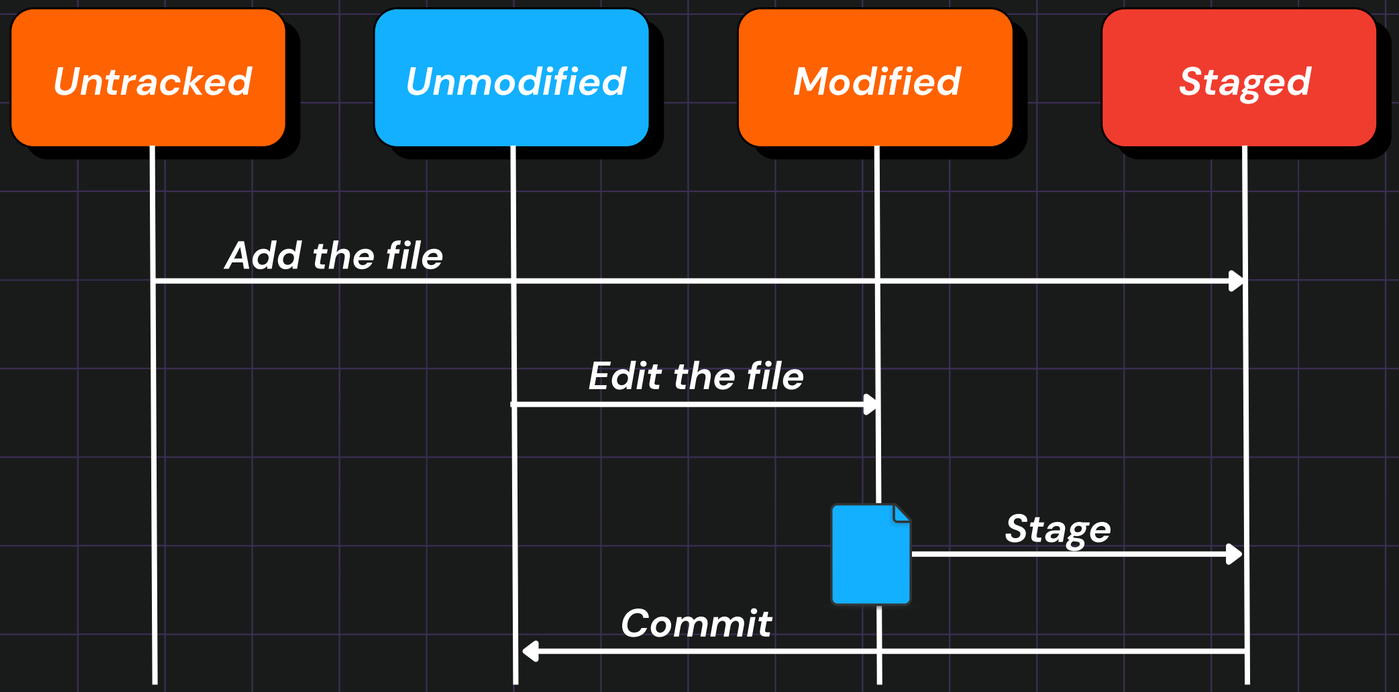
FILE STATES



File in VSCode

File in Git

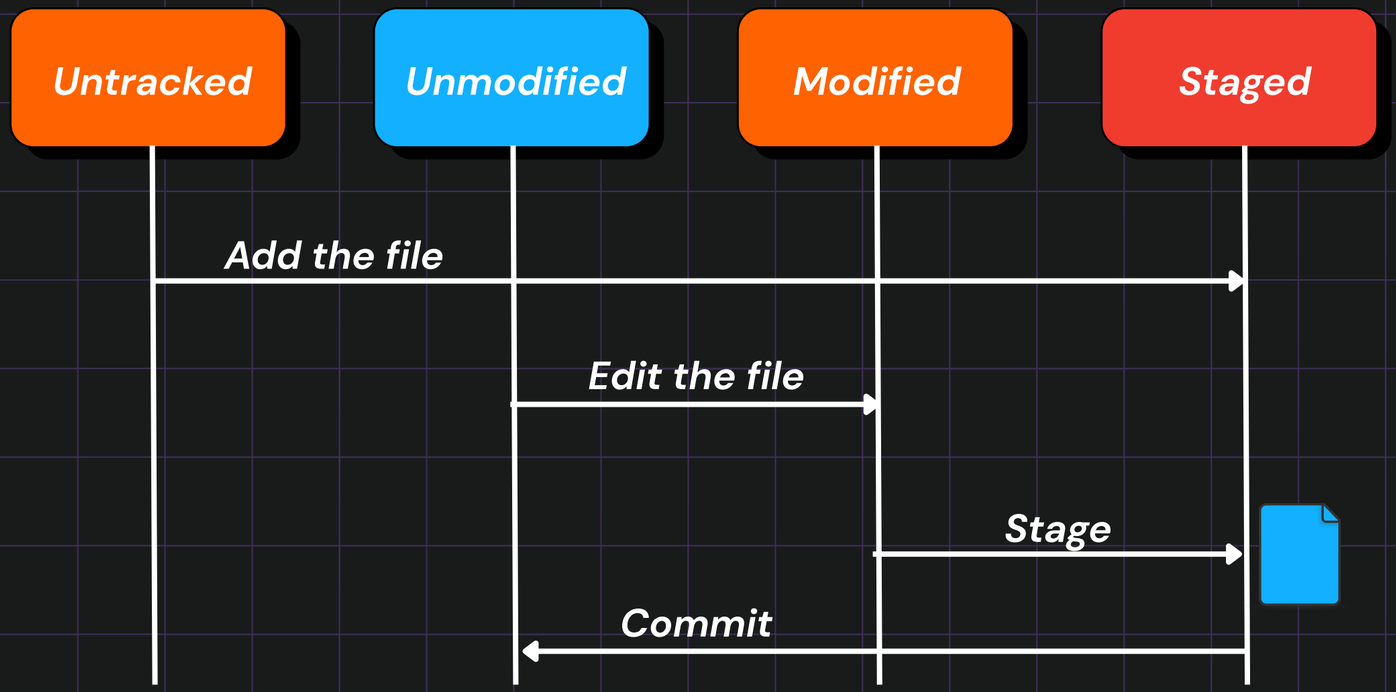
FILE STATES



File in VSCode

File in Git

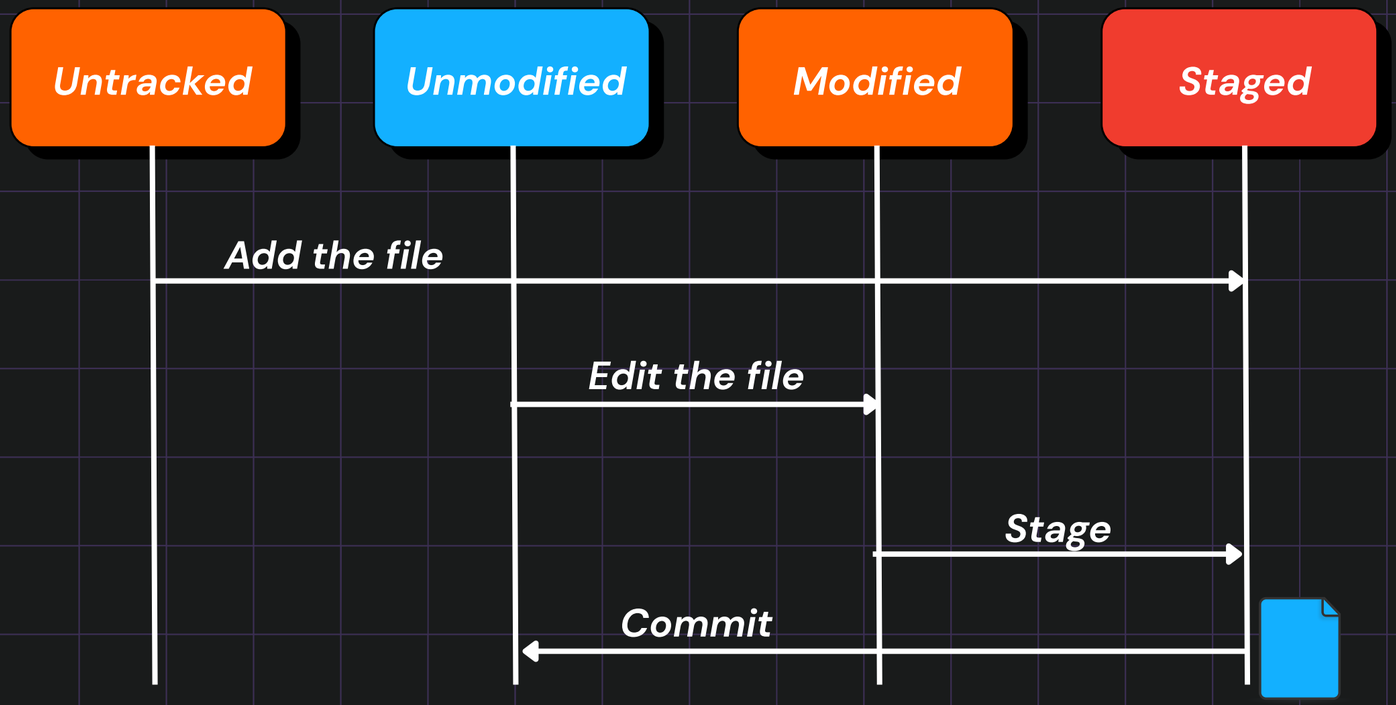
FILE STATES



File in VSCode

File in Git

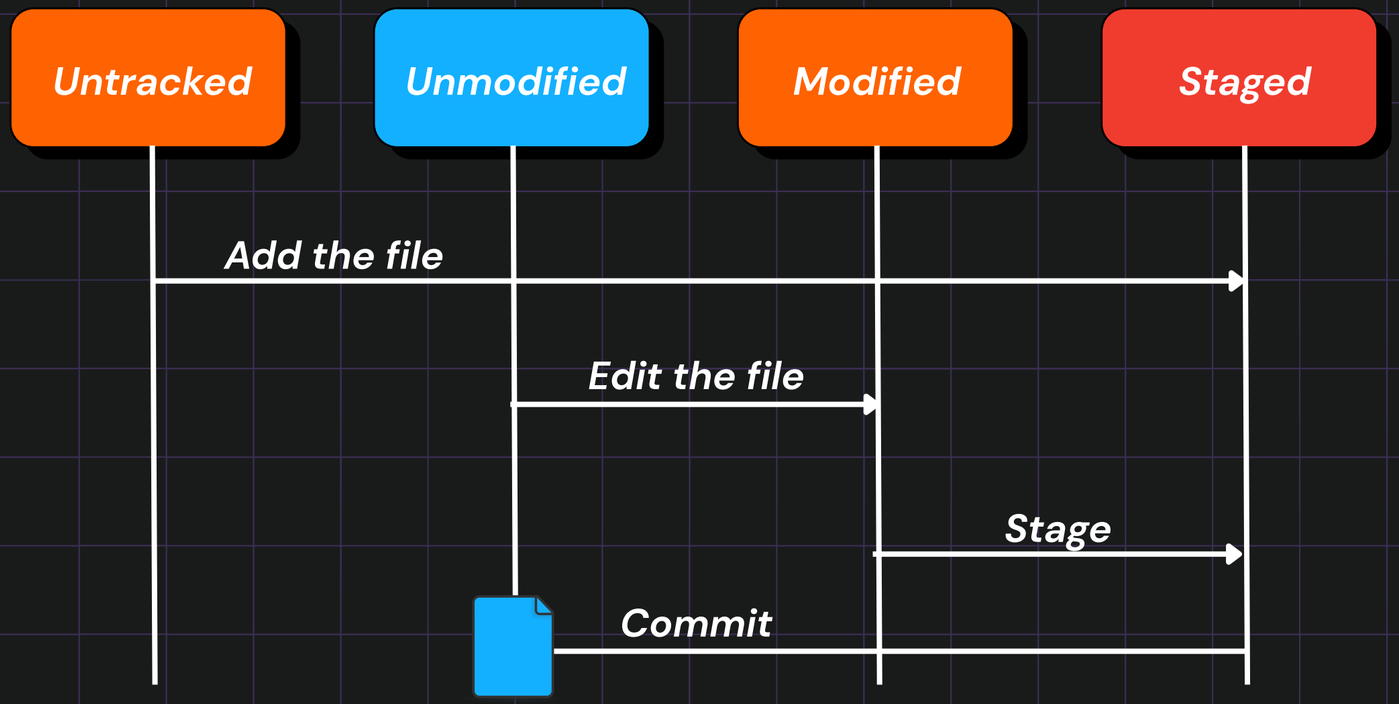
FILE STATES



File in VSCode

File in Git

FILE STATES

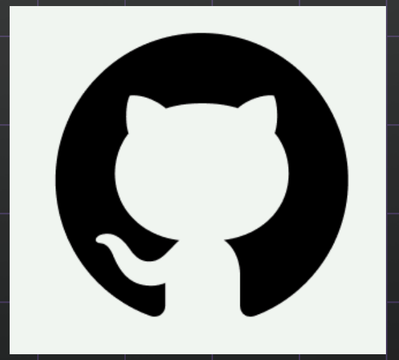


File in VSCode

File in Git

WLUG
COMMUNITY | KNOWLEDGE | SHARE

VS



WLUG
COMMUNITY | KNOWLEDGE | SHARE

GIT

- A free and Open Source VCS
- Works locally without internet connection
- Tracks changes and keeps history
- Works with any hosting platform

VS

GITHUB

- A proprietary website used to host Git Repos
- Needs internet connection since it is hosted on cloud
- Helps teams share and review work
- Just one of several Git repo hosting platforms

WLUG
COMMUNITY | KNOWLEDGE | SHARE

GITLAB

- Open Source Alternative to GitHub
- Repos can be self hosted
- DevOps features like CI/CD Pipelining and Container Registry

FEATURES

- Pull Requests: A way to ask project owner to review and merge your changes
- Fork: Making a copy of someone else's repo to your account so that you can make changes freely

WLUG
COMMUNITY | KNOWLEDGE | SHARE

BASIC LINUX COMMANDS

WLUG
COMMUNITY | KNOWLEDGE | SHARE

Print Working Directory

```
$ pwd
```

List Contents of Directory

```
$ ls
```

WLUG
COMMUNITY | KNOWLEDGE | SHARE

Creating a Directory

```
$ mkdir
```

Changing Directory

```
$ cd
```

WLUG
COMMUNITY | KNOWLEDGE | SHARE

Creating a File

```
$ touch
```

Viewing File Content

```
$ cat
```

WLUG
COMMUNITY | KNOWLEDGE | SHARE

Deleting a file

```
$ rm
```

WLUG
COMMUNITY | KNOWLEDGE | SHARE

TABLE OF CONTENT

- Installing & Configuring Git
- Basic Git Commands
- Git Branching & Merging
- Advance Git Commands
- Open Source Contribution

WLUG
COMMUNITY | KNOWLEDGE | SHARE

INSTALLING GIT

INSTALL COMMANDS

WLUG
COMMUNITY | KNOWLEDGE | SHARE

Install on Linux

```
$ sudo apt install git
```

WLUG
COMMUNITY | KNOWLEDGE | SHARE

CHECK THE VERSION

Check Git Version

```
$ git --version
```

WLUG
COMMUNITY | KNOWLEDGE | SHARE

CONFIGURING GIT

WLUG
COMMUNITY | KNOWLEDGE | SHARE

GIT CONFIGURATION

- Git needs to know who you are
- Name and email in every commit

SET UP GIT

Set your username

```
$ git config --global user.name "Your Name"
```

Set your email

```
$ git config --global user.email "you@example.com"
```

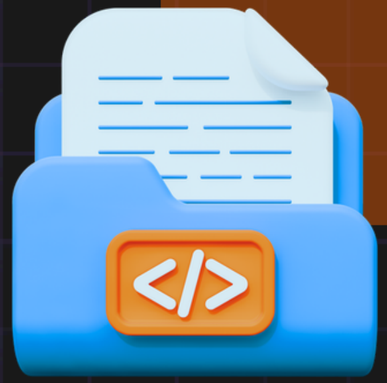
CHECK CONFIGURATION

Check what Git has saved

```
$ git config --list
```

WLUG
COMMUNITY | KNOWLEDGE | SHARE

BASIC GIT COMMANDS



INITIALIZE GIT

Initialize an empty repository

```
$ git init
```

CHECK THE GIT STATUS

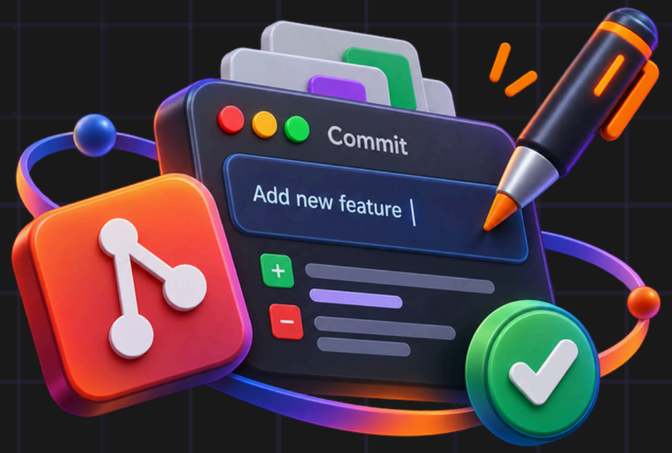
Check the status of a repository

```
$ git status
```

COMMIT CHANGES

Commit a message

```
$ git commit -m "message"
```



VIEW HISTORY

WLUG
COMMUNITY | KNOWLEDGE | SHARE

View Commit History

```
$ git log
```

WLUG
COMMUNITY | KNOWLEDGE | SHARE

.GITIGNORE

- .gitignore lists files to skip
- Skip passwords
- Only affects untracked files

WLUG
COMMUNITY | KNOWLEDGE | SHARE

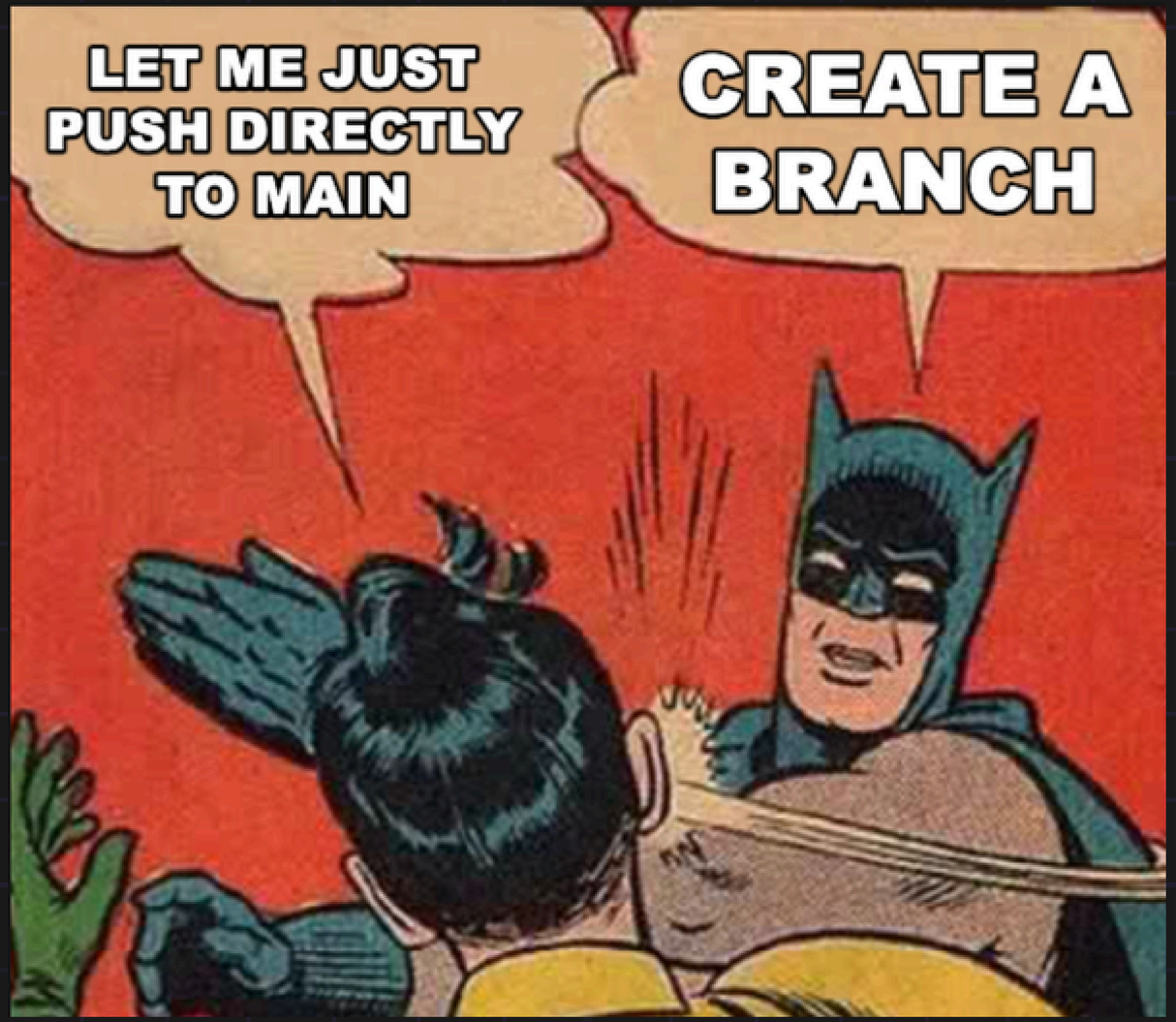
WLUG
COMMUNITY | KNOWLEDGE | SHARE

GIT BRANCHING

WLUG
COMMUNITY | KNOWLEDGE | SHARE

GIT BRANCHES

- A separate line of work
- Keep main clean, try new things
- Branches are quick and cheap

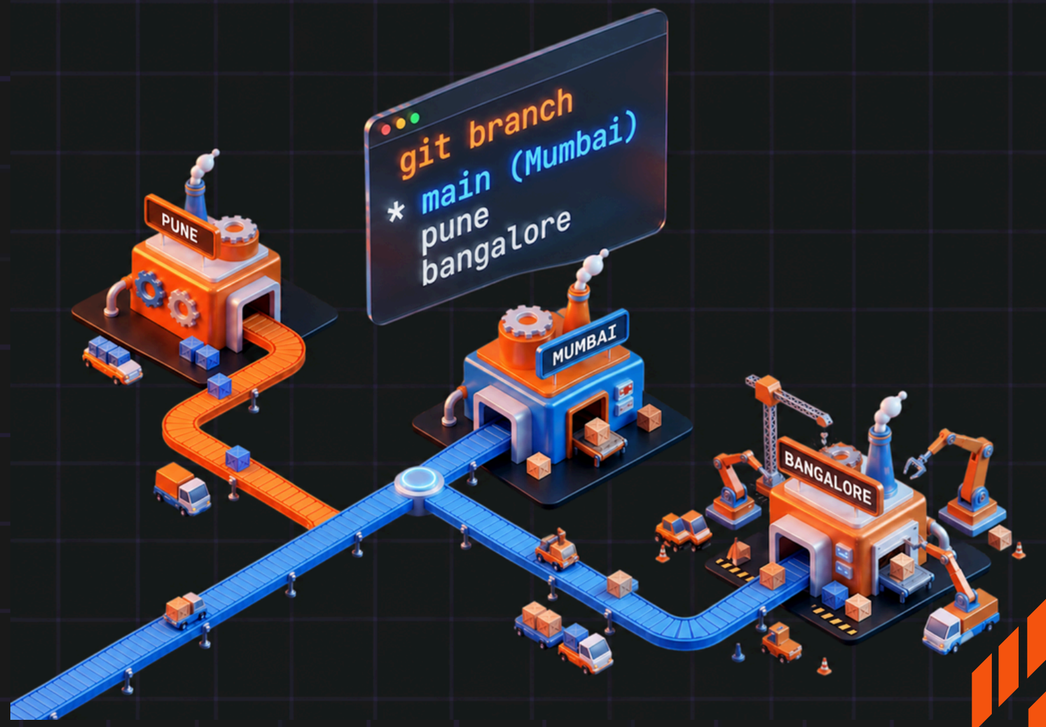

LET ME JUST PUSH DIRECTLY TO MAIN
CREATE A BRANCH

WLUG
COMMUNITY | KNOWLEDGE | SHARE

LIST ALL BRANCHES

Lists all local branches

```
$ git branch
```



CREATE BRANCH

WLUG
COMMUNITY | KNOWLEDGE | SHARE

Creates a new branch

```
$ git branch <branch_name>
```

RENAME BRANCH

Renames a branch

```
$ git branch -m <new name>
```

DELETE BRANCH

Deletes a branch

```
$ git branch -d <branch name>
```

WLUG
COMMUNITY | KNOWLEDGE | SHARE

SWITCH BRANCHES

Switch to another branch

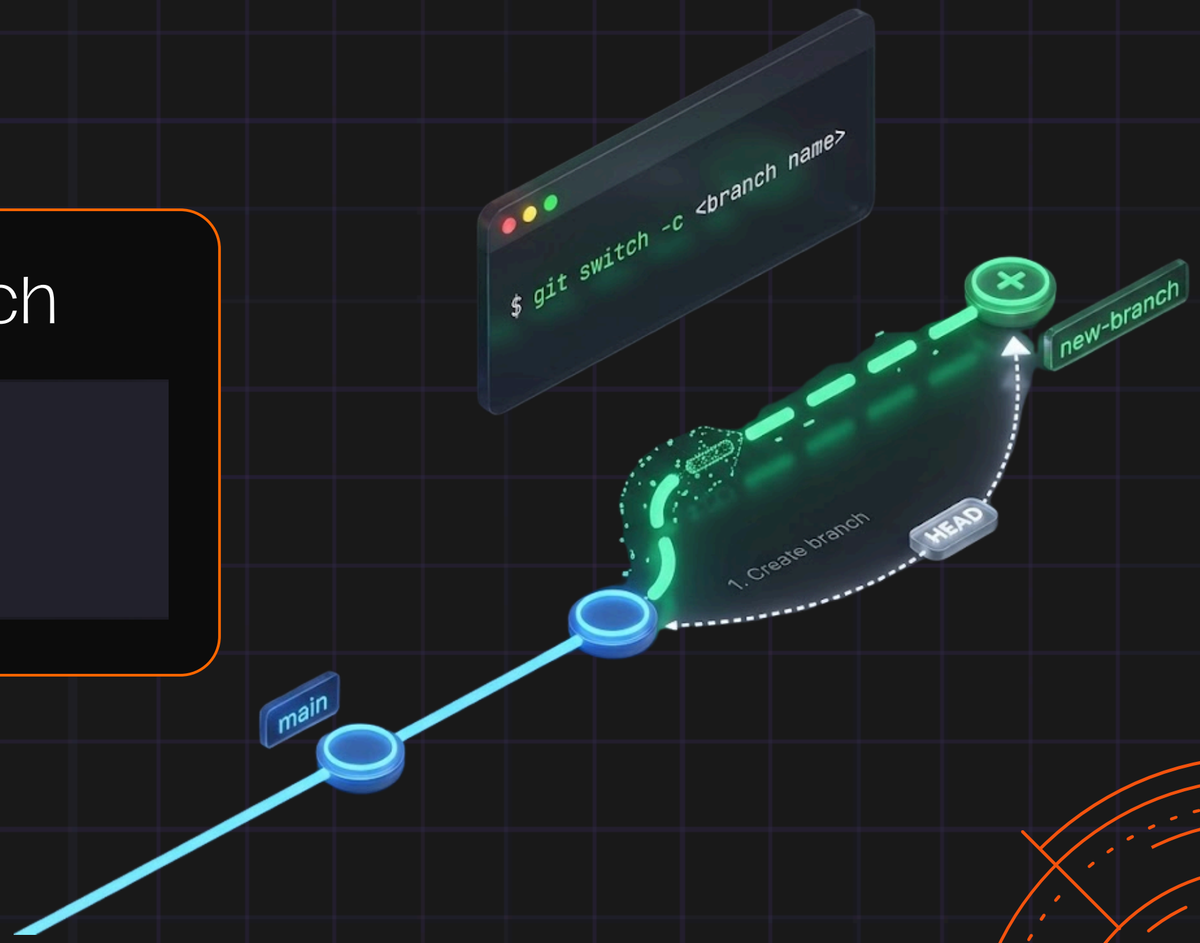
```
$ git switch <branch name>
```

CREATE AND SWITCH

WLUG
COMMUNITY | KNOWLEDGE | SHARE

Create and switch to a branch

```
$ git switch -c <branch name>
```



COMPARE BRANCHES

Compare between branches

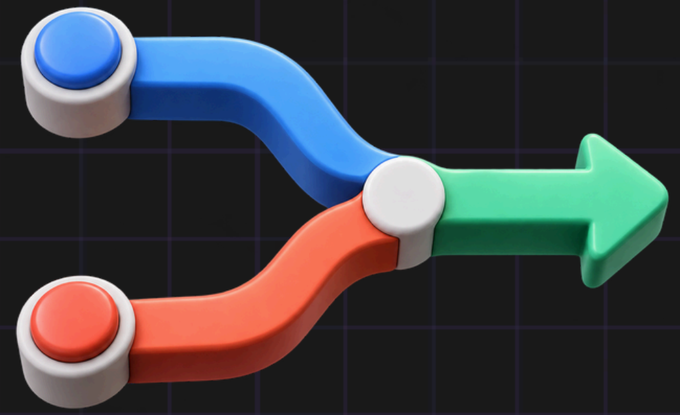
```
$ git diff <branch 1> <branch 2>
```

MERGE BRANCHES

WLUG
COMMUNITY | KNOWLEDGE | SHARE

Merges a branch on main

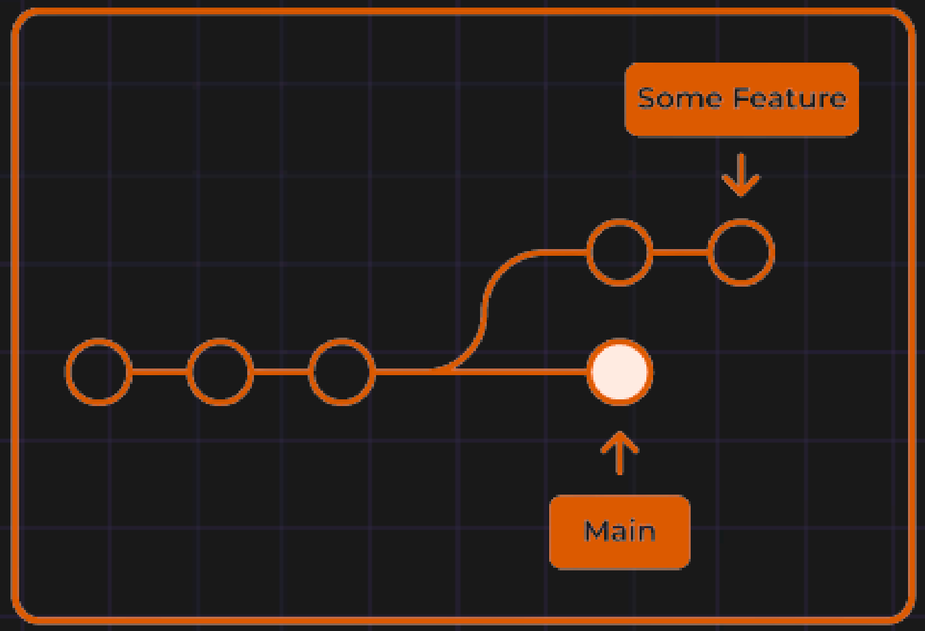
```
$ git merge <branch name>
```



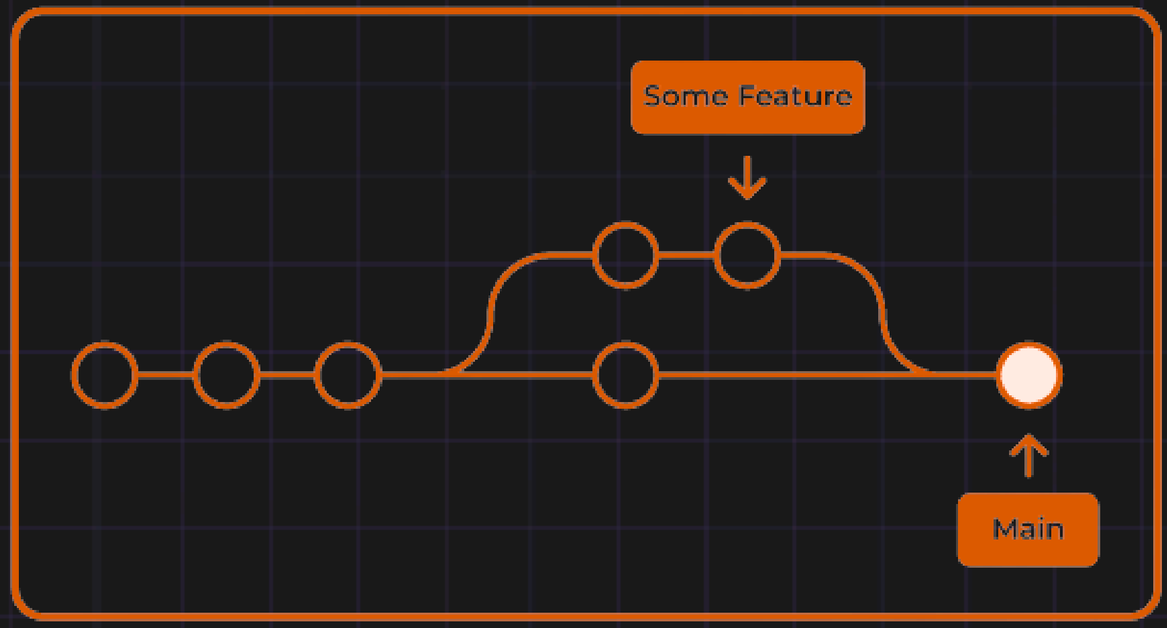
Git Merge

WLUG
COMMUNITY | KNOWLEDGE | SHARE

Before Merging

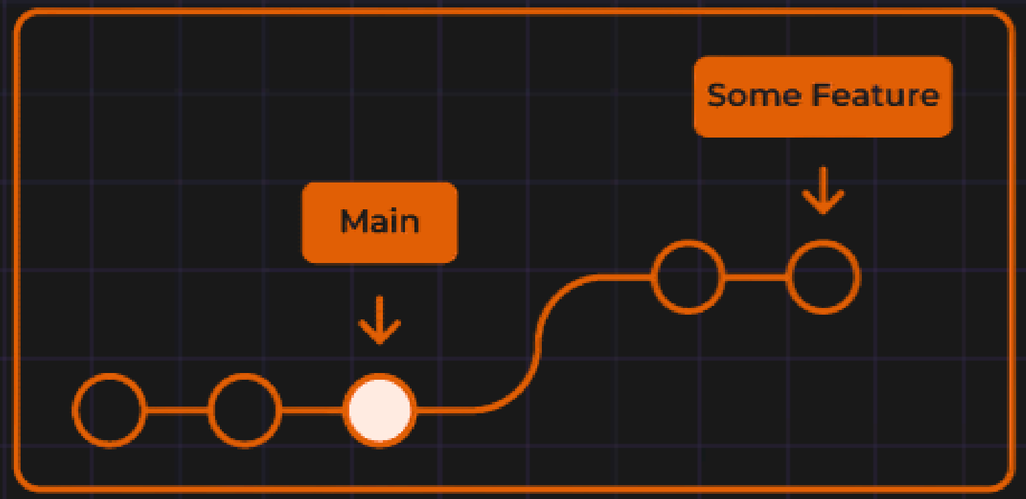


After a 3-way Merge

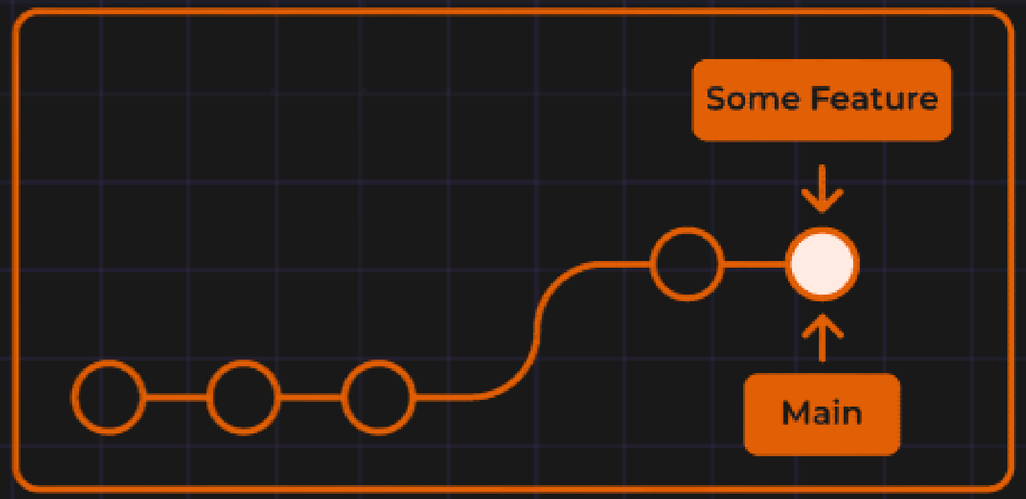


WLUG
COMMUNITY | KNOWLEDGE | SHARE

Before Merging



After a Fast-Forward Merge



WLUG
COMMUNITY | KNOWLEDGE | SHARE

3-WAY MERGE

VS

FAST FORWARD

- New merge commit
- Used when branches are diverged
- Merge commit has two parents
- Can cause merge conflicts

- No new commits
- Used when no divergence exists
- Just moves branch pointer
- Never causes merge conflicts

MERGE CONFLICTS

- Both branches edit the same line
- Contains conflict markers
- Fix by hand, then add and commit

WLUG
COMMUNITY | KNOWLEDGE | SHARE

CONFLICTS
CONFLICTS EVERYWHERE

WLUG
COMMUNITY | KNOWLEDGE | SHARE

ADVANCE GIT COMMANDS

GIT REBASE

- Move commits to another branch
- Tidies history before a merge
- Do not rebase pushed commits

REBASE COMMAND

Rebase onto main

```
$ git rebase main
```

WLUG
COMMUNITY | KNOWLEDGE | SHARE

MERGE

- Combine branches with merge commit
- Preserves original commit history
- Non linear commit graph
- Better for merging feature into main

VS

REBASE

- Adds commits on top of target branch
- Rewrites history (new commit hashes)
- Keeps commit history linear
- Used to clean commits before merging

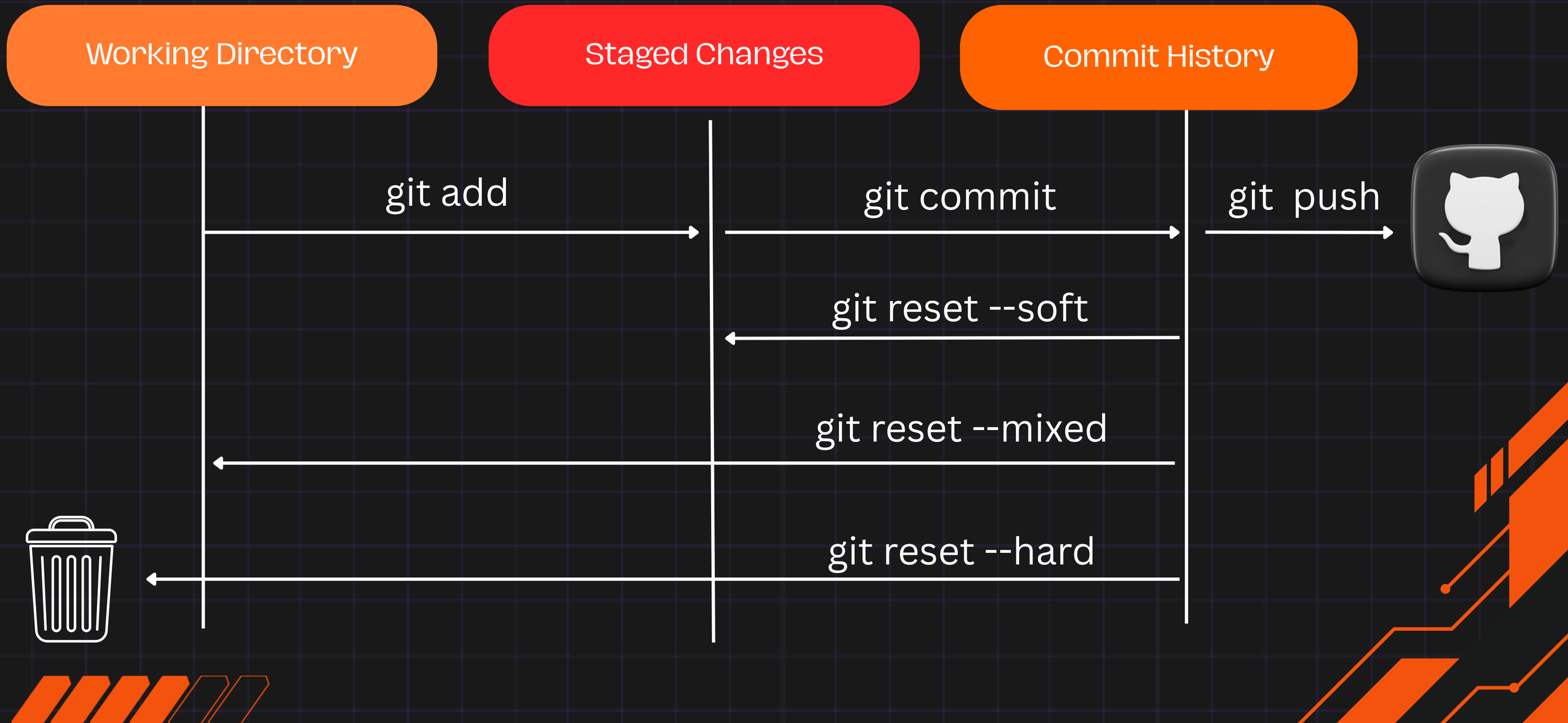
WLUG
COMMUNITY | KNOWLEDGE | SHARE

TIME MACHINE

GIT RESET

- Undo changes and modify history
- “ - - soft ” keeps your changes staged
- “ - - hard ” deletes them, be careful

WLUG
COMMUNITY | KNOWLEDGE | SHARE



RESET COMMANDS

Soft reset

```
$ git reset --soft HEAD~1
```

GIT REVERT

- Undoes one specific commit
- Adds a commit that undoes it
- History stays intact
- Safe on shared branches

REVERT THE COMMIT

Revert a specific commit

```
$ git revert <commit>
```

WLUG
COMMUNITY | KNOWLEDGE | SHARE

RESET

- Rewrites history
- Keep it local
- - soft keeps changes
- - hard deletes changes

VS

REVERT

- Adds a new commit
- History stays intact
- Safe when shared
- Undoes one commit

SUMMARY

- init, add, and commit to save your work
- Use branching and merging for parallel work
- Use rebase and revert to fix history

WLUG
COMMUNITY | KNOWLEDGE | SHARE

OPEN SOURCE CONTRIBUTION

WHAT IS A PULL REQUEST?

- A request to merge your changes
- Maintainers review and discuss it
- Approved code goes into main

WLUG
COMMUNITY | KNOWLEDGE | SHARE

START WITH GITHUB

- Sign up or sign in to GitHub
- Open the repo you want to contribute to

FORK THE REPOSITORY

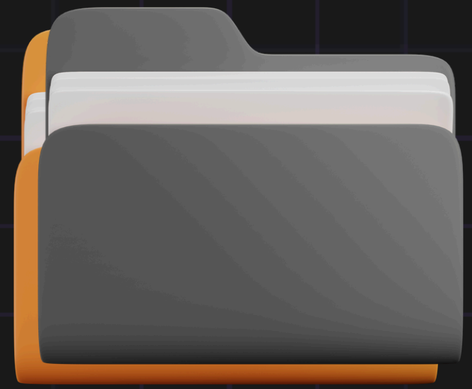
A fork is a copy of the repository on your GitHub

CLONE YOUR FORK

WLUG
COMMUNITY | KNOWLEDGE | SHARE

Clone your fork

```
$ git clone <your-fork-url>
```



COMMIT YOUR WORK

Commit your changes

```
$ git add .
$ git commit -m "message"
```

PUSH YOUR CHANGES

Push to GitHub

```
$ git push origin <branch_name>
```

WLUG
COMMUNITY | KNOWLEDGE | SHARE

CREATE A PULL REQUEST

- Open your fork on GitHub
- Click compare & pull request
- Describe your changes and submit

REVIEW AND MERGE

- Discuss and review the changes
- Fix bugs or update docs if asked
- Approved changes get merged

WAITING FOR A
PULL REQUEST
APPROVAL

WLUG
COMMUNITY | KNOWLEDGE | SHARE

THANK YOU